

Reconstruction of missing oceanic environmental field data using small-sample, multivariate methods

*Local Prior and Joint Spatiotemporal Learning

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Abstract—Marine monitoring records are often sparse in space and time and may be asynchronously missing across variables. We propose a small-sample multivariate reconstruction framework that uses a same-time local Gaussian prior and learns residual corrections from temporal states, spatial context, and cross-variable relations. The method is evaluated on nine marine regions with seven environmental variables using chronological train/validation/test splits and shared masks across all methods. At 70% random missingness, the proposed model is competitive with local interpolation, EOF-style reconstruction, and a joint convolutional recurrent baseline, while the best method changes with the variable and missing-data geometry. Ablation results identify the local prior as the most consistent information source and show that temporal and cross-variable branches provide variable-dependent corrections. The study therefore treats reconstruction as a conditional field-recovery problem and reports the operating conditions under which each information source is useful.

Index Terms—Marine environment; spatiotemporal data gaps; multivariate interpolation; DINEOF; optimal interpolation; evaluation protocol; applicable boundary

I. INTRODUCTION

Marine equipment is exposed to coupled marine and atmospheric conditions, including temperature, salinity, humidity, atmospheric SO₂, dissolved oxygen, and pH. These variables affect thermal loading, corrosion, coating degradation, and the construction of environmental load spectra. Monitoring records, however, are incomplete because of sparse trajectories, sensor outages, communication interruptions, and nonuniform sampling. The resulting task is conditional recovery of a multivariate field from partially observed space–time data, rather than unconstrained numerical filling.

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Optimal interpolation and EOF methods remain useful for small samples because they are inexpensive and interpretable [1]–[4]. Their covariance or low-rank assumptions can nevertheless oversmooth local gradients and do not directly represent nonlinear temporal or cross-variable dependencies. Deep models such as DINCAE, GRIN, MTGNN, and PriSTI provide richer representations [5]–[8], but their behavior depends strongly on sample size and missing-data geometry.

This paper addresses the resulting small-sample regime with a local-prior residual framework. A same-time Gaussian estimate supplies the observation-supported baseline; a ConvGRU state, spatial context, and learned seven-variable relations then predict a correction to that baseline, while visible measurements are copied unchanged. We evaluate this design under shared masks, strict chronological isolation, and several missingness geometries to identify which information sources are effective and when.

II. METHOD

A. Mechanism

As shown in Fig. 1, visible observations first define a same-time local Gaussian baseline. Temporal state, spatial context, and learned seven-variable relations then predict a residual correction, while observed entries are copied unchanged. The following equations define these operations.

- 1) **Sparse seven-variable field and mask.** The model receives a sparse seven-variable marine environmental field together with its visibility mask:

$$\mathbf{X}_t \in \mathbb{R}^{H \times W \times 7}, \quad \mathbf{M}_t \in \{0, 1\}^{H \times W \times 7}. \quad (1)$$

The seven variables are sea-surface temperature, air temperature, salinity, relative humidity, atmospheric SO₂ concentration, dissolved oxygen concentration, and pH.

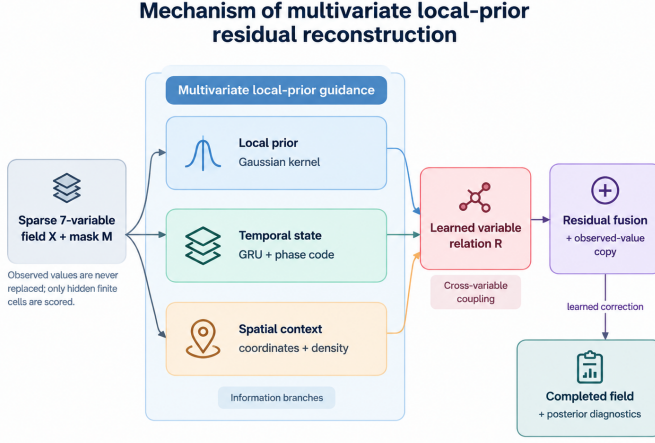


Fig. 1. Overview of the proposed reconstruction mechanism. Visible observations provide a local spatial prior; temporal, spatial-context, and variable-relation branches predict a residual correction, while observed values are copied unchanged.

- 2) **Local prior.** A spatial prior field is constructed from visible observations in the neighborhood of the same time step using a local Gaussian kernel:

$$\mathbf{P}_t(i, j) = \frac{\sum_{(u, v) \in \mathcal{N}(i, j)} K_\sigma(i - u, j - v) \mathbf{M}_t(u, v) \mathbf{X}_t(u, v)}{\sum_{(u, v) \in \mathcal{N}(i, j)} K_\sigma(i - u, j - v) \mathbf{M}_t(u, v) + \varepsilon} \quad (2)$$

where $\mathcal{N}(i, j)$ denotes the local neighborhood of position (i, j) , K_σ is the Gaussian kernel, and ε is a small stabilizing constant.

- 3) **Temporal state.** Historical feature states and the current temporal phase encoding are used to model the temporal evolution of the environmental field:

$$\mathbf{H}_t = \text{ConvGRU}(\mathbf{F}_t, \mathbf{H}_{t-1}, \phi_t), \quad (3)$$

where $\mathbf{F}_t$ denotes the current local features, $\mathbf{H}_{t-1}$ is the historical hidden state, and ϕ_t is the temporal phase encoding.

- 4) **Spatial context.** Geographic coordinates and local observation density are incorporated as spatial contextual information:

$$\mathbf{S}_t(i, j) = \mathbf{W}_2 \sigma(\mathbf{W}_1 [\lambda_i, \varphi_j, \rho_t(i, j)]^\top + \mathbf{b}_1) + \mathbf{b}_2, \quad (4)$$

where λ_i and φ_j denote longitude and latitude, respectively, and $\rho_t(i, j)$ represents the local observation density around position (i, j) .

- 5) **Learned variable relation.** A 1×1 convolution, or an equivalent channel-mixing layer, jointly learns the coupling relationships among the seven environmental variables:

$$\mathbf{V}_t = \text{Conv}_{1 \times 1}(\mathbf{X}_t, \mathbf{M}_t). \quad (5)$$

This branch represents statistical associations among temperature, salinity, humidity, dissolved oxygen, pH, and the other environmental variables.

- 6) **Residual fusion.** Local spatial priors, temporal states, spatial context, and learned variable relations are fused to predict a residual correction to the local prior:

$$\mathbf{R}_t = f_\theta(\mathbf{P}_t, \mathbf{H}_t, \mathbf{S}_t, \mathbf{V}_t), \quad (6)$$

$$\hat{\mathbf{X}}_t = \mathbf{P}_t + \mathbf{R}_t. \quad (7)$$

- 7) **Observed-value copy.** Original visible observations are preserved without modification, while only missing locations are replaced by model predictions:

$$\tilde{\mathbf{X}}_t = \mathbf{M}_t \odot \mathbf{X}_t + (1 - \mathbf{M}_t) \odot \hat{\mathbf{X}}_t, \quad (8)$$

where $\odot$ denotes element-wise multiplication.

- 8) **Completed field and diagnostics.** The model produces a completed seven-variable environmental field $\tilde{\mathbf{X}}_t$. The reconstruction is subsequently assessed using posterior diagnostics, including residual distributions, inter-variable correlations, spatial continuity, temporal continuity, and physical plausibility.

III. EXPERIMENTS

A. Dataset and Evaluation Protocol

The dataset contains nine marine regions, with 24 temporal fields collected for each region. Each field is represented on a common geographic grid and includes seven environmental variables: sea-surface temperature (tos , K), air temperature (tas , K), salinity (sos , psu), relative humidity (hurs , %), atmospheric SO_2 concentration (so2 , dataset units), dissolved oxygen concentration (o2 , mg L^{-1}), and pH (ph , dimensionless). The data were chronologically divided into 14 training fields, 5 validation fields, and 5 test fields for each region. No field from the validation or test periods was used for parameter fitting, normalization, or model selection.

To evaluate reconstruction under realistic monitoring conditions, missing values were generated by masking observed entries. The primary experiment used a 70% random missing rate. Additional random-missing experiments were performed with missing rates from 10% to 90%. For each missing rate, five shared mask seeds (20260906–20260910) were used for all methods, ensuring that the compared models received exactly the same visible observations and were evaluated on the same hidden entries. Missingness was sampled independently for different variables, so that the missing pattern of one environmental variable did not determine the pattern of another variable.

Four structured missing scenarios were additionally considered: continuous temporal gaps, grid-boundary missingness, asynchronous variable missingness, and missingness concentrated in the top 20% high-gradient areas. The grid-boundary scenario refers to the boundary of the gridded computational domain and is not interpreted as a physical coastline. These scenarios were used as stress tests for different observation geometries rather than as additional training data.

B. Training, Reconstruction, and Model Selection

For each training field, an internal self-supervised holdout was generated from the available observations. The model received the remaining visible values, the corresponding visibility mask, geographic coordinates, observation-density features, and temporal phase information. The loss was computed only on the artificially hidden target entries:

$$\mathcal{L}_{\text{train}} = \frac{1}{|\Omega_{\text{tar}}|} \sum_{(i,j,k) \in \Omega_{\text{tar}}} \left(\hat{X}_{t,i,j,k} - X_{t,i,j,k} \right)^2, \quad (9)$$

where Ω_{tar} denotes the target entries hidden for self-supervised training. Consequently, visible values were not treated as prediction targets, and the model could not use hidden target values as input.

The same training procedure was applied independently to each method. Model parameters were optimized using AdamW with a learning rate of 2×10^{-3} and a weight decay of 10^{-5} . The gradient norm was clipped to 1.0. Training was run for at most 55 epochs, with early stopping when the validation loss did not improve for 10 consecutive epochs. The model checkpoint with the lowest validation loss was retained for testing. All trainable models were implemented in PyTorch 2.11.0 and executed with CUDA 12.8 on an NVIDIA GeForce RTX 5060 Laptop GPU.

The mean and standard deviation used for normalization were estimated only from valid values in the 14 training fields. The same transformation was then applied to the validation and test fields, and all reported errors were converted back to the original physical units. During testing, model parameters were frozen. For each test field, only the observations that remained visible under the test mask were provided to the model. The hidden test values were never used for input construction, model selection, or parameter updates and were used only for final evaluation. The final completed field was obtained by copying visible observations directly and replacing only missing entries:

$$\tilde{\mathbf{X}}_t = \mathbf{M}_t \odot \mathbf{X}_t + (1 - \mathbf{M}_t) \odot \hat{\mathbf{X}}_t. \quad (10)$$

C. Compared Methods

The proposed method, denoted as OURS, was compared with four reference approaches under the same data split, input masks, target masks, and evaluation procedures:

- **Climatology**: reconstruction from the training-period climatological mean at each grid location and variable;
- **OI analogue**: local observation-based objective-analysis reconstruction using the nearest available observations;
- **EOF-style**: low-rank empirical orthogonal function reconstruction fitted using the incomplete training fields;
- **JointConvGRU**: a joint multi-variable convolutional recurrent network without the proposed local-prior residual formulation;
- **OURS**: the proposed model combining a local Gaussian spatial prior, convolutional feature extraction, ConvGRU temporal states, temporal phase encoding, coordinate

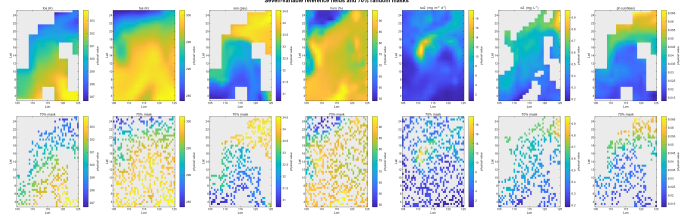


Fig. 2. Example of the seven-variable reconstruction input. The first row contains the original fields and the second row contains the independently masked observations supplied to the model. Each column corresponds to one environmental variable.

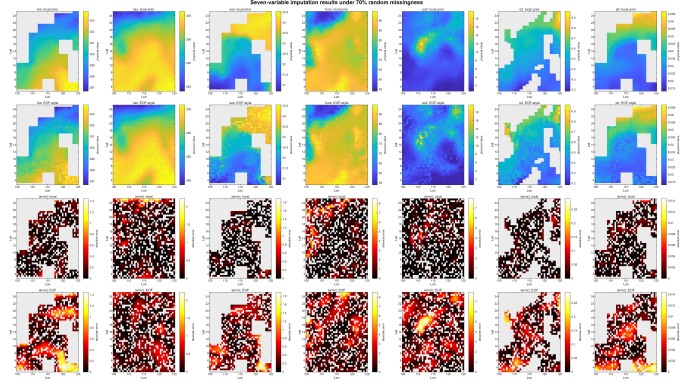


Fig. 3. Example reconstruction for the same test case. Completed fields are shown for all seven variables using the proposed method. Visible input values are copied unchanged, and only masked entries are replaced.

and observation-density features, learned seven-variable relations, and residual fusion.

All learning-based methods were trained using the same chronological split, self-supervised target construction, validation-based checkpoint selection, normalization procedure, and optimization budget. Non-learning reference methods were fitted using training-period data only and applied directly to the same masked validation and test fields.

D. Illustrative Multivariable Reconstructions

Fig. 2 shows the original and independently masked fields for one test case; Fig. 3 shows the corresponding OURS reconstruction with shared color limits within each variable.

IV. RESULTS

A. Main Results under 70% Random Missingness

Figure 4 presents the region-level RMSE under the primary 70% random-missingness condition. Errors are reported in the original physical units for each variable, and the error bars represent 95% intervals calculated with marine regions as the clustering units. Because the seven variables have different physical units and numerical scales, their RMSE values are compared only within each variable.

OURS achieved the lowest regional mean RMSE for four of the seven variables: 0.4271 K for air temperature (t_{as}), 0.1954 psu for salinity (s_{os}), 2.0745% for relative humidity (h_{urs}), and 0.4695 for the atmospheric SO_2 indicator.

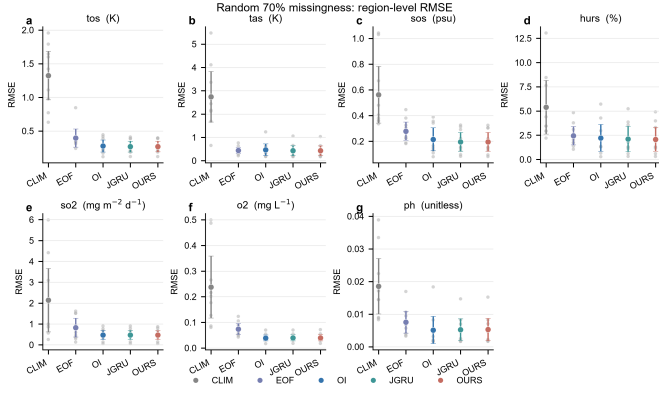


Fig. 4. Region-level RMSE under 70% random missingness. Each panel corresponds to one environmental variable. Gray points denote individual marine-region results, colored points denote regional means, and vertical bars indicate 95% intervals.

JointConvGRU obtained the lowest RMSE for sea-surface temperature (t_{os}), with an RMSE of 0.2676 K, whereas the RMSE of OURS was 0.2682 K. OI analogue performed best for dissolved oxygen (o_2) and pH, with RMSE values of 0.0392 mg L^{-1} and 0.0051, respectively. The differences between OURS and the two strongest reference methods were generally small for t_{as} , sos , so_2 , and t_{os} .

Confidence intervals overlap for most variables, indicating variable-specific rather than universal gains.

B. Sensitivity to the Random Missingness Rate

Figure 5 shows RMSE as the random missingness rate increases from 10% to 90%. Error generally grows with missingness, with larger changes for $hurs$, t_{as} , and t_{os} .

OURS remained the lowest-error method for $hurs$ across the evaluated missingness rates. It was also competitive for sos , t_{as} , and so_2 , particularly at low and intermediate missingness rates. At 90% missingness, JointConvGRU achieved lower RMSE for o_2 and sos , EOF-style achieved the lowest RMSE for t_{as} , and OI analogue was best for so_2 and pH. JointConvGRU generally performed best for t_{os} up to 70% missingness, while OURS became comparable or slightly better at 90%. These crossings indicate that the relative ranking changes as the amount of same-time spatial support decreases.

C. Performance under Structured Missingness

Figure 6 summarizes the lowest-error method for each variable under four structured missingness conditions: continuous temporal gaps, grid domain-edge missingness, asynchronous variable missingness, and missingness concentrated in the top 20% highest-gradient cells. The figure reports categorical winners for each variable.

The winning method changes across temporal gaps, domain edges, asynchronous variables, and high-gradient cells. OI and EOF remain strong for broad or temporally supported fields, while OURS is competitive for selected variables.

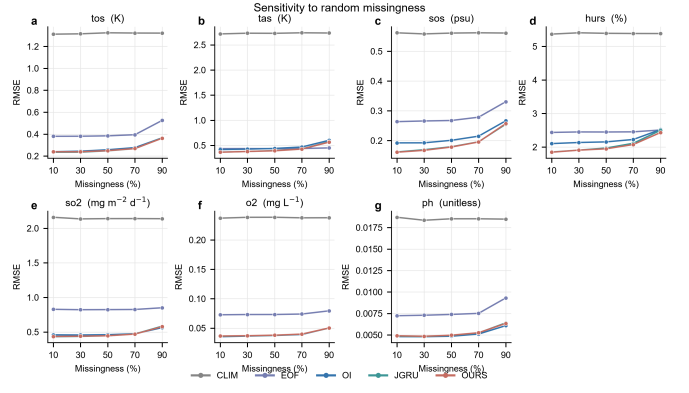


Fig. 5. Sensitivity of reconstruction RMSE to the random missingness rate. Results are shown separately for the seven environmental variables in their original units.

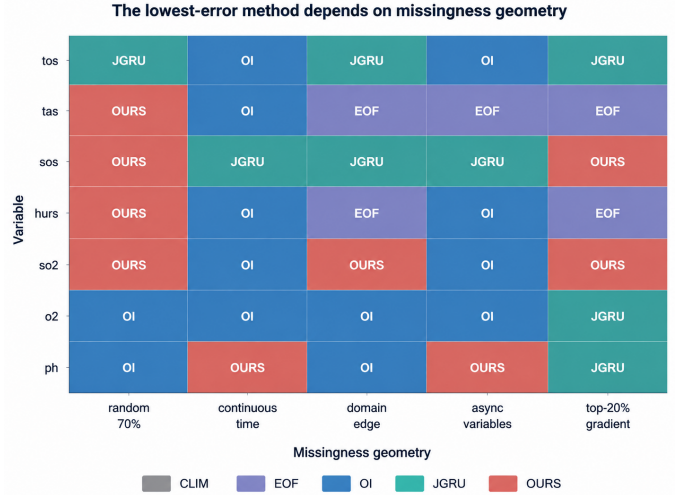


Fig. 6. Lowest-error method for each environmental variable under five missingness geometries. The random-70% column corresponds to the primary experiment; the remaining columns represent structured stress tests.

D. Component Ablation

Figure 7 reports the percentage change in RMSE after removing one component from OURS under a fixed training and masking seed.

Removing the local-neighborhood branch increased the RMSE for sos , t_{as} , so_2 , and o_2 by approximately 5.4%, 3.1%, 6.8%, and 5.9%, respectively. This branch therefore provided the most consistent contribution in the tested configuration. The effects of removing the variable-relation, coordinate, temporal-phase, and observation-density branches depended on the target variable. For example, removing the coordinate branch reduced the pH RMSE by 6.1% but increased the $hurs$ RMSE by 3.6%; removing the density branch increased the pH RMSE by 5.9% but reduced the $hurs$ RMSE by 2.6%. These mixed directions indicate that the additional branches provide variable-dependent corrections rather than uniform gains.

OURS contains 108,176 trainable parameters versus

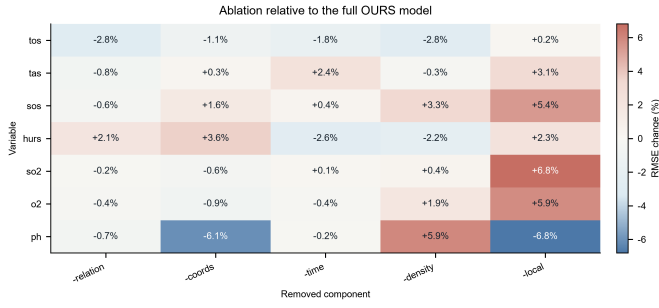


Fig. 7. Component ablation of OURS relative to the complete model. Values show the percentage change in RMSE after removing one component. Positive values indicate increased error.

106,327 for JointConvGRU.

V. CONCLUSION

This paper presented a small-sample framework for reconstructing multivariate marine environmental fields from partially observed space-time records. A same-time local Gaussian prior supplies the observation-supported baseline, while temporal state, spatial context, and learned seven-variable relations predict residual corrections; visible measurements are copied unchanged. Under shared random and structured masks, the best method depended on the variable and missing-data geometry. The local prior was the most consistent component in ablation, whereas temporal and cross-variable branches provided variable-dependent supplements. These results support conditional field recovery as a practical formulation for sparse environmental monitoring. Future work should evaluate larger observation archives and incorporate external circulation and coastline information.

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